

Comprehensive Metal Ecology — Analysis Overview

MacGregor & Arkin (2026) · 9 analytical layers · Thick borders = central narrative (Layers 1 → 2 → 4 → 6 → 9)

Layer 1 Core Hypothesis & Primary Test	Primary PGLS (H1 confirmed) Metal-gene density → cross-biome niche breadth $\beta = -0.021$ · SE = 0.0037 · $p = 2 \times 10^{-9}$ · n = 1,574 Pagel's $\lambda = 0.757$ · $\Delta AIC = -29.4$ · $r^2 = 0.046$ Pre-registered; direction confirmed on GTDB r214	Archaeal replication n = 95 · $\beta = -0.014$ · p = 0.119 Directionally consistent with H1 Non-significant: underpowered (~40% power) Expanded archaeal dataset: future priority	Australia NGSa null test Soil metal conc. → niche breadth: $\beta = -0.002$, p = 0.755 Predictor insensitivity, not population failure Soil conc. ≠ genomic KO density (different predictor) Per-metal Cu/Zn: FDR q = 0.041 (pH-controlled)	Confounder checks (5 pre-specified) Genome size: 47% attenuation → robust (p = 0.006) GC content: 24% attenuation → robust Latitude: amplified (−52%) → suppressor variable Biome (6%) · isolation source (15%) → robust
Layer 2 Internal Functional Split	Resistance / Detoxification (106 KOs) $\beta = +0.003$ · p = 0.656 · null on cross-biome breadth Positive on geochemical breadth ($\beta = +0.073^{**}$) Positive on co-occurrence degree ($\beta = +4.1^{***}$) HGT-mobile; decoupled from genome streamlining	Cofactor biosynthesis (7 KOs) $\beta = -0.033$ · p = 5×10^{-9} · strongest subcategory Fe-S cluster · molybdopterin · cobalamin Constitutively expressed; vertically inherited Equal to housekeeping streamlining baseline	Split-magnitude permutation ($\Delta\beta = 0.035$) 1,000 random partitions of 140-KO set (106 vs 7) $\Delta\beta = 0.035259$ exceeds ALL null splits · p < 0.001 −6 SD above null median (SD = 0.007) ABC / AMR / TCS splits all smaller ($\Delta\beta \leq 0.032$)	Non-metal cofactor & cofactor jackknife Non-metal cofactor: $\beta = -0.029$ = metal $\beta = -0.033$ → Biosynthetic cost drives signal, not metal identity Jackknife (leave-1-KO-out): all 3-KO remainders significant β range −0.016 to −0.029 · no sign changes
Layer 3 Multi-Axis Niche Framework	Cross-biome Levins' B (primary axis) Cofactor: $\beta = -0.033^{***}$ · Resistance: $\beta = 0$ (null) Transport/homeostasis: $\beta = -0.022^{**}$ Sensing/regulation: $\beta = -0.018^{**}$ · Non-metal cofactor: $\beta = -0.029^{***}$ Polarity confirmed: cofactor specialist = generalist metabolically	Geochemical breadth (Env PC1) Within-genus SD across pH, temp., 5 GeoROC metals Resistance: $\beta > 0$ (positive) → wide chemical niche Cofactor: null on geochemical breadth Polarity inverted: resistance expands local niche only	Social niche breadth (co-occurrence degree) Phi-coefficient network · 162,022 soil samples, 3,149 genera Pagel's $\lambda = 0.48$ –0.57 (phylogenetically informative) Resistance: $\beta = +4.1^{***}$ · metal-rich genera are hubs 56% top-quartile partners also metal-rich; Firmicutes-biased	Scale boundary: CWM & within-soil Within-soil niche breadth (β soil): null signal Community-weighted mean (3 metal sources, 7 covariates): $ \beta < 0.006$ · all p > 0.2 · cross-biome phenomenon only Metal-gene / niche link undetectable at within-biome scale
Layer 4 Mechanism: Two-Scale Phylo-D	Two-scale phylogenetic signal (275 KOs) Genus-level Pagel's λ on continuous KO presence fractions Genome-level Fritz & Purvis D on binary presence (18,961 tips) Spearman $\rho = -0.041$ · p = 0.49 → orthogonal measures Capture distinct evolutionary processes (vertical vs HGT)	13 double-signal KOs ($D > 0.2$, $\lambda < 0.3$) Dispersed at BOTH evolutionary scales → repeated HGT Hg resistance: merD ($\lambda = 0.165$) · merE ($\lambda = 0.102$) As resistance: aoxB · gesA · gesB (gold/tellurite) Cu resistance: cusA $\lambda = 0.79$ (vertically inherited; negative control)		Differential mobility mechanism Cofactor (Fe-S, molybdopterin, cobalamin): high λ + low D → vertically inherited; metabolically essential Resistance: low λ + high D → HGT-mobile; tracks local metal Pangenome coreness: cofactor 2.2× more prevalent than resistance
Layer 5 Functional Generalisation	KEGG functional landscape (19 categories) 14/19 significant (BH-FDR q < 0.05) · Metal ranks 11th Streamlining baseline: $\beta = -0.030$ to -0.035 (housekeeping) 5 confirmed true-negative controls: ABC, AMR, motility, TCS, glycan Appropriate null = housekeeping baseline, not zero	Expanded KEGG cofactor set (47 KOs) 8 KEGG modules: M00880 · M00121/926 · M00122/924 M00846 · M00175/176 (Fe-S cluster assembly) Full set: $\beta = -0.011$ · p = 0.010 Cobalamin (18 KOs) primary driver: $\beta = -0.011$ · p = 0.005	Non-exclusive KO classification (93 KOs) KOs assigned proportionally across functional categories Metal-dependent metabolism expanded: 1 → 93 KOs Still patterns with resistance ($\beta = 0$), not cofactor Constitutive biosynthetic coupling is the key discriminant	Comparison gene families (ABC, AMR, TCS) ABC transporters: Lipid/LPS export only shows cofactor-like β AMR: all positive ($\beta = +0.018$ for efflux pumps) TCS: all positive · metal split is family-specific Requires constitutive metabolic coupling to target phenotype
Layer 6 External Validation	Frossard 2017 (Hg-contaminated soils, Switzerland) Industrial Hg contamination · bacterial niche breadth Hg-tolerant: B_std = 0.407 vs non-tolerant: 0.233 · p < 0.0001 Effect size comparable to primary finding Independent replication: direction confirmed	Li 2022 (Hg field gradient, China) Long-term Hg-contaminated agricultural soils CWM niche breadth elevated: 0.365 vs control 0.233 Community-level corroboration of genus-level pattern Direction consistent with H1 prediction	NGSA geochemical replication (soil Cu, Zn) NGSA ICP-MS metals · AMI 165 · 200 km join · n = 482 Cu: $\beta = -0.011$ · FDR q = 0.041 · Zn: $\beta = -0.011$ · q = 0.041 Pb: q = 0.057 · Ni: q = 0.061 · Co: reversed ($\beta > 0$) Partial replication (2/5 metals) · lower $\lambda = 0.32$ –0.35	Mechanism boundaries (Goff 2024 · Frossard 2018) merA + merR on conjugative plasmids at Oak Ridge Reservation Abdelmageed 2021: anoxic specialists via hgcA Hg methylation Frossard 2018 field gradient: null · microcosm: null → Boundary: aerobic tolerance ≠ anaerobic methylator specialism
Layer 7 Robustness & Sensitivity	8 pre-specified sensitivity analyses 7/8 directionally consistent (all $\beta < 0$ except raw count) OLS: $\beta = -0.032$ · Brownian: $\beta = -0.018$ Northern hemisphere: $\beta = -0.030$ · soil-restricted: $\beta = -0.033$ Raw count (no per-Mb normalisation): NS → confirms per-Mb required	MAG quality & phylum stratification High-quality MAGs ($\geq 90\%$ comp., $\leq 5\%$ contam., n = 511): $\beta = -0.018$ · p = 0.005 → signal not driven by low-quality MAGs Proteobacteria (n = 677) + Actinobacteria (n = 204): FDR significant Firmicutes borderline · Bacteroidetes high $\lambda = 0.917$ (near-Brownian)	Cofactor-housekeeping residualised (H4c) ρ (expanded cofactor density, translation density) = 0.038 ≈ 0 Residualised $\beta = -0.013$ · p = 0.004 (stronger than naive) Semi-partial R ² : cofactor = 0.008 · genome size = 0.010 Cofactor signal is independent of general metabolic investment	Niche-breadth metric & species-level Bootstrap (100 resamples): r = 0.999 vs original · $\Delta\beta = 0.00083$ EMP validation (n = 539): $\beta = -0.019$ · p = 0.211 (p < 10^{-6}) Species-level (5 genera): 5/5 negative · 3/5 significant (p < 0.001) Coreness-matched permutation: emp p = 0.298 (landscape artefact ruled out)
Layer 8 CatBoost LOPO ML Validation	CatBoost LOPO · 15 environmental responses 11 phyla (≥ 10 genera) · Leave-One-Phylum-Out cross-validation 13/15 responses: positive average Spearman ρ Soil pH: $\rho = 0.213$ · CSU-As: $\rho = 0.174$ · Bedrock Co: $\rho = 0.163$ Complements PGLS: non-linear signal, cross-phylum transferability	SHAP feature importance hemH + cobT: porphyrin/cobalamin biosynthesis cluster cusA (K07787): most consistent across PGLS and CatBoost emrB (K03446): highest mean [SHAP] across env. responses copA (K17686): specialist pattern · negative on metal axes	KO prioritisation (composite ranking) hemH (K01772) · ferroxidase · porphyrin pathway cobT (K09883) · cobalamin adenosyltransferase cusC (K07796) · Cu/Ag efflux outer-membrane · bmrR (K19575) zur (K09823) · zinc uptake regulator · spans both sides of split	
Layer 9 Experimental Proposal	Cobalamin co-culture experiment Producer:auxotroph ratio varied in replicated gradient chambers Predict: community breadth ≠ cobalamin biosynthesis capacity Producers: P. stutzeri · S. mellotti (complete pathway) cobT knockout in producer: within-strain experimental control	HGT mechanism validation Resistance/niche assoc. weaker in HGT-acquired vs. vertical genera? Reconciliation-based gene-tree / species-tree analyses Plasmid localisation (PLSDB / PlasClass) across high-quality assemblies Synteny analyses of resistance gene neighbourhoods	Conjugation microcosm & KO follow-up Generalist vs specialist Burkholderiales + mer plasmid Fitness cost assay · transfer rate · retention without selection Transcriptomics across metal gradients 5 prioritised KOs for functional follow-up (hemH-cobT-cusC-bmrR-zur)	

Thick border = central narrative box

Thin border = supporting analysis